B.TECH DATA SCIENCE

Module 3: Data Collection, Cleaning and Exploration

B.Tech Data Science — Expanded Study Notes and Exam Preparation

Contents
Four chapters of expanded concepts, definitions, examples, applications, methods and revision questions.
Includes side headings on every study page and important questions at the end for exam preparation.

Prepared for learning and exam revision. Follow your university syllabus and faculty guidance where they differ.

1.1 Planning Data Collection

Study focus: Data Collection, Cleaning and Exploration. Read the side headings, understand the example, and revise the key ideas.

Purpose

Data collection obtains observations needed to answer a defined question. Before gathering data, clarify the decision, target population, variables, time period and quality requirements.

Target population

The target population is the full group to which the study intends to relate. It may be all enrolled students, all transactions in a period or all sensors in a system.

Variables and units

Define each variable, unit of measurement, acceptable range and collection frequency. Ambiguous definitions can make otherwise accurate records incomparable.

Sampling plan

Choose a sampling method that fits the question and resources. Record how units were selected, likely sources of bias and how non-response will be handled.

Permissions and security

Confirm permission, consent and access requirements before collecting personal or restricted data. Collect only what is necessary and define how long it will be retained.

1.2 Data Quality Dimensions

Study focus: Data Collection, Cleaning and Exploration. Read the side headings, understand the example, and revise the key ideas.

Accuracy

Accuracy concerns whether a value correctly represents what it claims to measure. An incorrect sensor reading or mistyped age reduces accuracy.

Completeness

Completeness concerns whether required values are present. Missing values may be concentrated in particular groups, so a missing-value count alone may not tell the whole story.

Consistency

Consistency means values and definitions agree across records and sources. Dates, category names and units should use standard conventions.

Validity

Validity concerns whether values satisfy defined rules, such as a percentage between 0 and 100 or a date in an allowed period.

Uniqueness and timeliness

Uniqueness checks unintended duplicate records. Timeliness checks whether the data is recent enough for the intended task. Quality dimensions should be measured and documented.

1.3 Data Sources and Acquisition

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Databases and files

Data may be acquired from relational databases, CSV files, spreadsheets, JSON documents, application programming interfaces or data warehouses. Each source has different schema and access considerations.

Surveys and interviews

Surveys can gather standardised responses, while interviews provide more detailed context. Question wording, response options and non-response can introduce bias.

Sensors and logs

Sensors and software logs can generate high-frequency observations. Check timestamps, calibration, device identifiers, outages and whether recorded events match real-world events.

Public datasets

Public datasets can support learning and research, but check the licence, collection date, definitions, population coverage and documentation before using them.

Acquisition record

Record the source, owner, retrieval date, licence, schema, data version and transformations. This creates provenance and helps reproduce results.

1.4 Sampling, Bias and Data Leakage

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Random sampling

Random sampling uses a random process to select observations. Simple random, stratified and cluster sampling are different designs with different strengths.

Selection bias

Selection bias occurs when the selection process systematically excludes or over-represents some members. A voluntary online poll may not represent the whole population.

Measurement bias

Measurement bias occurs when a measurement method systematically misrepresents a quantity. Examples include a poorly calibrated sensor or a survey question that leads respondents.

Data leakage

Data leakage occurs when training or analysis uses information that would not legitimately be available at the time of prediction or evaluation. It can produce overly optimistic results.

Prevention

Define prediction timing, split data before fitting transformations, avoid future-derived variables and check duplicates or related records across train and test sets.

2.1 Data Inspection and Profiling

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First inspection

Review row and column counts, column names, data types, example values, missingness, unique values and summary statistics. Inspecting a small sample can reveal format problems.

Type checks

A number stored as text may not support arithmetic; a date stored inconsistently may sort incorrectly. Convert types only after understanding the original representation.

Range checks

Use domain rules to flag impossible or suspicious values. For example, a percentage outside 0–100 may be invalid, but a value outside the usual range is not automatically wrong.

Duplicate checks

Identify exact duplicate rows and duplicate business keys. Some repeated records may be valid events, so use the meaning of a row to decide what counts as a duplicate.

Profiling report

Summarise quality issues, their frequency, likely causes and proposed handling. Keep the original data and record each change.

2.2 Missing Values

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Why values are missing

Values may be missing because a question was skipped, a device failed, a field did not apply or a record was lost. Understanding the reason helps choose a suitable treatment.

Deletion

Rows or columns may be removed when missingness is limited and deletion will not seriously distort the analysis. Removing many rows can reduce power and create selection bias.

Simple imputation

Numerical values may be imputed using a mean or median, and categories using a mode or explicit unknown label. Simple imputation is easy but can understate uncertainty or distort relationships.

Model-based methods

More advanced imputation estimates missing values using other variables or multiple plausible datasets. Such methods need assumptions, validation and careful reporting.

Missingness indicators

A separate indicator can record whether a value was missing. It may preserve useful information about the data collection process, but it should be used thoughtfully.

Documentation

Record the number of missing values, the chosen method, reasons where known and the impact on analysis. Compare results where the treatment choice could change conclusions.

2.3 Duplicates, Inconsistent Values and Dates

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Exact duplicates

Exact duplicate rows can arise from repeated imports or logging errors. Identify them using all fields or a defined unique key before deciding whether removal is appropriate.

Near duplicates

Near duplicates may differ in spelling, punctuation or formatting. Name matching and record linkage can help, but automated matching may combine different people or entities incorrectly.

Category standardisation

Values such as “Andhra Pradesh”, “AP” and “A.P.” may refer to the same category. Create documented mapping rules rather than changing labels informally.

Dates and time zones

Dates should use a consistent format. Time-zone differences, daylight-saving changes and ambiguous day/month ordering can create errors in temporal analysis.

Validation after cleaning

Recheck row counts, unique keys, value ranges and summary statistics after changes. Keep a log so the process can be reviewed or reversed.

2.4 Outliers and Noisy Data

Study focus: Data Collection, Cleaning and Exploration. Read the side headings, understand the example, and revise the key ideas.

Definition

An outlier is an observation unusually far from the bulk of the data under a chosen measure. Noise is unwanted variation or error in measurements.

Possible causes

Outliers may result from data entry errors, sensor faults, mixed populations or genuine rare events. The cause determines whether correction, exclusion or retention is appropriate.

Detection methods

Use box plots, scatter plots, z-scores, robust statistics and domain thresholds. Each method has assumptions and may behave differently for skewed distributions.

Treatment choices

Possible actions include correcting a verified error, retaining a valid rare event, applying a justified transformation, using robust methods or conducting sensitivity analysis.

Avoid automatic removal

Deleting every extreme value can remove important evidence and bias results. Record the rationale and compare conclusions with and without questionable observations when appropriate.

3.1 Exploratory Data Analysis (EDA)

Study focus: Data Collection, Cleaning and Exploration. Read the side headings, understand the example, and revise the key ideas.

Meaning and purpose

Exploratory data analysis uses summaries and visualisations to understand a dataset before formal modelling. It helps reveal distributions, relationships, unusual records and questions for further investigation.

Univariate analysis

Univariate analysis studies one variable at a time. Numerical summaries and histograms help describe centre, spread, skew and potential outliers.

Bivariate analysis

Bivariate analysis examines two variables, such as sales by price or attendance by score. The appropriate method depends on whether the variables are numerical or categorical.

Multivariate analysis

Multivariate exploration examines several variables together. Grouped charts, correlation matrices and carefully selected plots may reveal interactions or hidden structure.

EDA limits

Patterns found during exploration may be accidental. Confirm important findings on new data or with a suitable analysis, and avoid treating visual association as proof of causation.

3.2 Charts for Different Data Types

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Bar chart

Bar charts compare counts or summaries across categories. Bars are separated to show distinct groups, and category labels should be readable.

Histogram

A histogram displays the distribution of a numerical variable using adjacent bins. Bin width affects the visible shape, so conclusions should not rely on one arbitrary setting.

Box plot

A box plot shows quartiles, median and potential outliers. It is useful for comparing distributions across groups, although it hides some detail about the distribution shape.

Scatter plot

A scatter plot shows the relationship between two numerical variables. Look for direction, form, clusters and outliers, while remembering that correlation does not establish causation.

Line chart and map

Line charts are useful for ordered time trends. Maps can show geographic patterns when location is meaningful and geographic scales are not misleading.

Chart design

Use a clear title, labelled axes, units, readable scales and accessible colours. Avoid unnecessary 3D effects and truncated axes that exaggerate differences.

3.3 Summary Statistics and Relationships

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Centre and spread

Mean and median describe centre; range, IQR, variance and standard deviation describe different aspects of spread. Select summaries that fit the distribution and measurement scale.

Skewness

A skewed distribution has an asymmetric tail. In a right-skewed distribution, a few high values may pull the mean above the median; robust summaries may be more representative.

Correlation

Correlation summarises association between numerical variables. A near-zero linear correlation does not rule out a nonlinear relationship.

Group comparisons

Compare groups using counts, proportions, means or medians as appropriate. Check group sizes and uncertainty; a large difference in a tiny group may be unstable.

Simpson's paradox

An overall relationship can differ from relationships within subgroups when group composition differs. Explore relevant variables before drawing a broad conclusion.

3.4 Feature Engineering and Encoding

Study focus: Data Collection, Cleaning and Exploration. Read the side headings, understand the example, and revise the key ideas.

Feature engineering

Feature engineering creates useful input variables from raw data using domain knowledge and the task definition. Examples include day of week from a date or duration from start and end times.

One-hot encoding

One-hot encoding creates indicator columns for categories. It avoids imposing an artificial numerical order on nominal categories, though it can create many columns for high-cardinality fields.

Ordinal encoding

Ordinal encoding maps ordered categories to numerical ranks when their order is meaningful. Applying arbitrary ranks to nominal categories can mislead some models.

Date and text features

Dates can provide month, weekday or elapsed-time features. Text may be transformed into tokens or numerical representations depending on the task and privacy requirements.

Leakage caution

Feature creation must use only information available at prediction time. Feature definitions and fitting procedures should be learned from training data where appropriate.

4.1 Scaling and Transformation

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Standardisation

Standardisation commonly computes z = (x – mean)/standard deviation. It centres a variable and scales it by its standard deviation, which can help methods sensitive to feature scale.

Min-max scaling

Min-max scaling often maps values into [0,1] using (x–minimum)/(maximum–minimum). New values outside the training range may fall outside that interval unless handled explicitly.

Log transformations

A log transformation can reduce right skew and compress large values. It requires attention to zero and negative values and changes the interpretation of differences.

Robust scaling

Robust scaling uses statistics such as median and IQR, making it less affected by extreme observations than mean-standard-deviation scaling.

Correct workflow

Split data before fitting preprocessing steps. Fit scaling parameters on training data, then apply the learned transformation to validation, test and future data.

4.2 SQL for Data Preparation

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SELECT and WHERE

SELECT chooses columns; WHERE filters rows. Clear queries should select only needed fields and make filter conditions explicit.

ORDER BY and LIMIT

ORDER BY sorts results according to specified columns. LIMIT restricts the number of returned rows in systems that support it, which is useful for inspection.

GROUP BY and aggregates

GROUP BY forms groups for aggregate calculations. Common functions include COUNT, SUM, AVG, MIN and MAX. Distinguish row counts from counts of non-null values.

JOIN operations

Joins combine related tables using keys. Inner joins keep matching records; left joins retain all rows from the left table and add matching right-side values where available.

Data checks in SQL

Queries can identify nulls, duplicates, invalid categories and inconsistent totals. Check whether joins unexpectedly multiply rows because keys are not unique.

4.3 Data Pipelines, Provenance and Reproducibility

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Pipeline stages

A data pipeline moves information through ingestion, validation, transformation, storage and delivery. Each stage should have clear inputs, outputs and failure handling.

Validation rules

Automated rules can check schema, allowed ranges, required fields, uniqueness and freshness. Failed checks should trigger a visible alert rather than silently passing bad data.

Provenance

Provenance documents source systems and transformations. It helps explain why a value changed and allows teams to trace a report back to the records used.

Reproducibility

Version data definitions, code and configuration where possible. Record the date, source version and transformation logic used to produce an analysis.

Operational monitoring

Monitor pipeline failures, processing delays, row counts and changes in data distributions. A pipeline that runs successfully may still deliver incorrect data, so quality checks remain essential.

5. Important Exam Questions and Revision

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Long-answer questions

1. Explain data quality dimensions. 2. Describe stages of data cleaning. 3. Explain EDA and suitable charts for different data types. 4. Discuss methods for handling missing values and outliers. 5. Explain feature encoding and scaling. 6. What is data leakage and how is it prevented?

Short-answer questions

1. Define data dictionary. 2. What is data provenance? 3. Differentiate completeness and accuracy. 4. Define imputation. 5. What is a box plot? 6. State the purpose of one-hot encoding. 7. What is a SQL join?

Practical questions

1. Suggest cleaning steps for a table containing duplicate IDs and missing ages. 2. Choose a chart to compare monthly sales. 3. Explain how to investigate an extreme sensor reading. 4. Write SQL logic to group records by category and count them.

Revision checklist

Review data acquisition, sampling bias, quality dimensions, missing data, duplicates, outliers, EDA, chart selection, encoding, scaling, SQL and data provenance.

Exam tip

Organise answers with headings such as definition, causes, methods, example and limitations. For cleaning questions, state why each action is justified and how you would validate the result.